

P R E S E N T A T I O N

Brain Tumor Classification using MRI





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Abstraction



Brain tumors are life-threatening conditions that require early and accurate diagnosis for effective treatment.

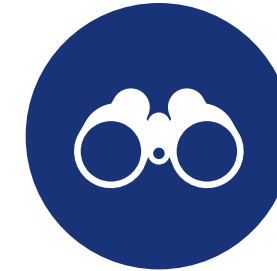
The Brain Tumor Diagnosis System leverages advanced artificial intelligence and medical image processing to detect and classify brain tumors using **MRI or CT scan images**.

This system employs deep learning techniques to automate tumor identification, distinguishing between benign and malignant cases with high precision. By integrating AI-driven analysis with a user-friendly interface, the system enhances diagnostic accuracy, reduces human error, and accelerates decision-making for medical professionals.

The goal is to improve early detection rates, optimize treatment planning, and ultimately enhance patient outcomes in neurology and oncology.



Data description



- This dataset contains **7023 images** of human brain MRI images which are classified into 4 classes: glioma - meningioma - no tumor and pituitary.
It divide to **5712 training** and **1311 testing**
- Data link
https://www.kaggle.com/datasets/masoudnic_kparvar/brain-tumor-mri-dataset



What model is used?



We utilized the **MobileNet** model instead of a traditional Convolutional Neural Network (CNN). MobileNet is a lightweight deep learning model optimized for efficiency while maintaining **high accuracy (91.6%)** (as Xception Model). Compared to standard CNNs(**84.1%**), MobileNet provided higher accuracy in brain tumor detection while requiring fewer computational resources. Its depthwise separable convolutions significantly reduce model size and processing time, making it ideal for real-time and mobile applications. This choice allowed us to achieve precise tumor classification with improved performance.



Applications



The Brain Tumor Diagnosis System can be applied in various medical and research fields to enhance early detection, diagnosis, and treatment planning. Some key applications include:



Telemedicine

Enables remote diagnosis and second opinions for better accessibility.



Hospitals and Clinics

Assists radiologists, neurosurgeons, and oncologists in tumor detection, classification



Personalized Treatment

Helps doctors track tumor progression and optimize treatment plans.



Colab Notebook Link



<https://colab.research.google.com/drive/1eoPGX2PPWzKqaIMWa02BkqdGRra7ZlIB?usp=sharing>



Conclusion



Our project successfully implemented a Brain Tumor Diagnosis System using the MobileNet model, achieving higher accuracy than traditional CNNs. By leveraging deep learning and medical image processing.

The system enhances early tumor detection and classification, aiding medical professionals in making faster and more reliable diagnoses.

The lightweight architecture of MobileNet ensures efficiency, making the system suitable for real-time applications.

This project demonstrates the potential of AI in healthcare, improving diagnostic precision and contributing to better patient outcomes.



Our Team



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Thank You.

For Your Attention